

HEXCRUNCH

Game Design Document

Core Gameplay, Player Experience, Board Logic, Cascade System and Design Direction

Current design state: Core gameplay complete; scoring, progression, win conditions and board orchestration remain open.

This document describes HexCrunch as a game, not as a software implementation. It distinguishes established design from open design space.

1. Game Overview

HexCrunch is a mobile puzzle game built around a large hexagonal board formed by concentric rings of hexagonal tiles. The player navigates a persistent two-tile selection — an inner-ring Actor and an adjacent outer-ring Follower — and swaps their colors. Every valid connection between neighboring rings can be selected, except swaps between tiles of the same color, which are skipped because they would have no gameplay value.

The game is designed for extremely low-effort one-thumb play. The player can keep the thumb in a comfortable resting area, use short directional swipes to move the selection, and tap anywhere to perform a swap. The player does not need to reach for individual tiles or cover the board with the hand.

The board itself becomes part of the tactile experience. The active selection is kept in the lower part of the large hexagon. When navigation would take the selection too far to the left or right, the board rotates and repositions underneath the selection. The result is that the player feels as though they are turning and manipulating a large mechanical hexagonal object while making very small thumb movements.

The main pattern-making objective in the current core design is to create a complete monochromatic ring. When an entire ring becomes one color, that ring clears and the outer structure crunches inward. Tiles from outer rings are pulled into empty inner positions through the board's actual neighbor relationships. This inward movement can itself create additional completed rings, enabling chain reactions and cascades comparable in emotional payoff to established match-3 games, but with a radial and hexagonal character.

The present prototype contains the complete core interaction, ring-completion rule, inward crunching and cascade-capable resolution. It does not yet contain the higher-level systems that shape what board states appear next: refill orchestration, difficulty progression, scoring, win/loss structure, level goals, or meta progression. Those are the next major design frontier.

2. High-Level Design Pillars

- One-thumb immediacy — the game should be physically effortless to control.
- Simple input, deeper consequence — navigation and swapping are easy; deciding what to swap should carry the puzzle depth.
- The board is an object — rotation, snap, motion and crunching should make the large hexagon feel tangible and playful.
- Readable geometry — the fixed ring structure should become understandable through visual language rather than heavy explanation.
- Radial cascades — successful moves should be capable of triggering satisfying inward chain reactions.

- Deterministic interaction, controlled future — the player's available actions should be clear and fair, while the game may intelligently orchestrate future states to preserve fun, challenge and reward.
- No unnecessary input — invalid or meaningless actions, such as swapping identical colors, should be removed from the player's workload.

3. Core Game Concept

3.1 The Board

The playfield is a single large hexagon composed of multiple concentric hexagonal rings. The geometry remains fixed throughout play. Board positions do not shift structurally; what changes is the color occupying each position.

Each playable ring contains six more positions than the previous ring. In the current five-ring setup, the ring capacities are 6, 12, 18, 24 and 30 tiles, for a total of 90 active colored positions.

An additional outer boundary ring exists only as structural helper geometry. It marks the full stop of the board and does not participate in normal gameplay, crunching, refilling or movement.

3.2 Board Adjacency

A tile belongs to one ring and is connected in two meaningful ways:

- Tangentially: to the previous and next tile around the same ring.
- Radially: to one or more neighboring tiles on the next inner or next outer ring.

Corner positions and side positions differ in radial connectivity. A corner tile can connect to three positions on the next outer ring, while a non-corner side tile connects to two. This asymmetry is part of the natural hexagonal geometry and is fundamental to the puzzle rather than an exception.

3.3 Permanent Positions, Moving Colors

A board position is permanent. A colored game piece is conceptually the color/content occupying that position. When two tiles are swapped, the positions themselves do not change; the colors exchange positions. Animations are used to make this feel like physical pieces crossing or moving.

4. Player Controls and Navigation

4.1 Input Vocabulary

The intended mobile input vocabulary is deliberately minimal:

- Swipe Left / Right — move through valid Actor–Follower pairings around the neighboring rings.
- Swipe Up / Down — change the active inner ring and corresponding outer-ring relationship.
- Single Tap — swap the currently selected Actor and Follower.

A tap can be made anywhere. The player is not required to tap the selected tiles themselves.

4.2 Actor and Follower

The active selection is always a pair across two neighboring rings:

- Actor — the selected tile on the inner ring.
- Follower — a directly adjacent selectable tile on the next outer ring.

The Actor always lives on the inner of the two active rings, and the Follower always lives on the outer ring. Together they define the current swap edge.

4.3 Complete Pair Traversal

Every geometrically valid Actor–Follower combination between two neighboring rings can be reached by navigation. Horizontal movement does not merely move from tile to tile. It walks through all selectable radial relationships: the player can move through one Actor's possible outer neighbors and then continue seamlessly to the next Actor and its outer-neighbor relationships.

This means the fundamental selectable object is effectively the connection between two adjacent-ring tiles, not a single tile in isolation.

4.4 Same-Color Pairs

Swapping equal colors would create no change, so same-color Actor–Follower pairs are considered invalid for swapping. Navigation automatically moves past them rather than requiring the player to stop, tap and receive no result. Such positions may be visually marked as blocked during traversal.

4.5 Repeated Move-and-Swap Convenience

If the player repeatedly moves in one direction and taps to swap, the game can recognize the pattern and omit repeated directional swipes. After enough repeated move–swap actions, continued taps can advance the selection and swap automatically in the same direction. This is a convenience feature rather than a central game rule.

4.6 One-Thumb Ergonomics

The control scheme is intended to let the player's thumb remain near its natural resting area. Unlike direct-manipulation puzzle games, the player does not need to reach across a dense board, accurately hit small tiles, or obscure the puzzle while interacting.

The screen can therefore remain visually readable during play, and the board can support a relatively dense arrangement without making input correspondingly difficult.

5. Board Motion and the Physical Feel of Play

The active selection is visually kept in the lower portion of the large hexagon. The practical lower interaction sector is bounded by downward-sloping edges on the left and right. When horizontal

navigation would push the selection beyond this comfortable region, the entire board rotates and repositions rather than forcing the player's selection and thumb to chase around the screen.

This gives the game an important tactile illusion: the player does not simply move a cursor over a static puzzle; the player feels as though they are turning the large hexagonal object itself.

The rotation is not a change to game geometry. It is a visual orientation correction and an interaction aid. The logical ring and neighbor relationships remain unchanged.

6. Visual Language of the Board

HexCrunch uses tile states to communicate the puzzle's structure without requiring the player to directly select individual cells.

Visual State	Player-Facing Meaning
Idle	A normal active colored tile with no special selection role.
Ring Face	Marks the ring context associated with the current Actor selection.
Actor	The inner-ring tile currently selected as the primary swap tile.
Follower	The outer-ring tile currently paired with the Actor and available to swap.
Blocked	A relationship encountered during navigation that cannot be swapped because both tiles already have the same color.
Stray	An empty/off tile: the board slot still exists visually, but no color is currently assigned. It is a placeholder awaiting a new color and is not an active gameplay cog in that moment.

Arrows are used around the current selection to communicate continuation and direction through the ring relationships. Selection animation, scale changes, links, background emphasis and board rotation are intended to reinforce the feeling of a responsive physical object.

7. Color Economy and Initial Board Inventory

The current core design uses five colors. At the start of a game, the total population of those colors is distributed in counts of 6, 12, 18, 24 and 30. Which specific color receives which population is randomized.

These five quantities exactly match the five playable ring capacities. Therefore, the board begins with a finite color inventory whose population structure corresponds mathematically to the ring structure.

The current prototype preserves the total population of each color across clearing and respawning. Completing a ring does not permanently destroy that color from the game; its pieces can return to the refill pool.

This makes HexCrunch fundamentally a finite redistribution/sorting system rather than an infinite stream of unrelated random pieces.

8. Core Player Action: Swap

A swap exchanges the colors of the currently selected Actor and Follower. The visual presentation makes the two pieces appear to move into one another's positions, but the board's underlying geometry remains fixed.

After the swap resolves, the game checks whether a complete monochromatic ring has been created. If not, control returns to normal navigation. If a ring has been created, the player interaction pauses and the crunch/cascade resolution system takes over.

9. Ring Completion

9.1 Ring-Making Rule

A ring is complete when every active position in that ring contains the same non-empty color. Partial runs, lines or clusters are not enough; the entire ring must become monochromatic.

This makes the unit of success geometric and whole-board-oriented. The player is not simply matching three local pieces; they are assembling a complete circular structure.

9.2 Resolution Priority

When multiple completed conditions exist, the current resolution behavior checks rings from the inside outward and begins resolving the first completed ring found. Further completed rings can be discovered as the crunch continues.

10. The Crunch

Completing a ring causes it to clear. Visually, the completed ring is emphasized and collapses away. The positions remain part of the board, but their colors become empty.

This creates a band of vacancies. Instead of applying conventional downward gravity, HexCrunch resolves those vacancies radially: colors from outer neighboring rings are pulled inward through valid hexagonal adjacency relationships.

The result is an inward collapse wave. The geometry of the board makes the cascade visually distinct from traditional match-3 gravity while serving a similar emotional function.

10.1 Pulling Rules

Each empty inner position can only pull from valid adjacent positions on the next outer ring. Several empty positions may compete for the same outer tiles, so the game must resolve claims without allowing one outer tile to fill multiple vacancies.

The current crunch logic also favors coherent color movement. For a set of vacancies, it evaluates available outer colors and tries to assign the largest possible group of compatible neighboring tiles of the same color before moving on to the remaining vacancies.

If multiple colors can produce equally large valid groups, one of those best options may be selected. This gives the crunch a tendency to preserve or strengthen color structure instead of behaving like purely arbitrary gravity.

10.2 Multi-Pass Collapse

The inward pull is repeated in passes. After outer tiles move inward, new vacancies appear farther from the center. The process continues until no more outer tiles can be pulled into available inner positions.

During this process, empty positions may be shown as Stray placeholders.

11. Cascades and Chain Reactions

After the inward pull settles, the game checks again for complete monochromatic rings. If the crunch has created another complete ring, that ring clears and another inward collapse begins before the board is fully refilled.

This creates HexCrunch's cascade system. One intentional player swap can cause:

1. A ring to complete.
2. That ring to clear.
3. Outer colors to crunch inward.
4. The new arrangement to complete another ring.
5. A second clear and crunch to begin.
6. The sequence to continue until no additional ring completes.

This cascade potential is intended to deliver the same class of pleasurable chain-reaction feedback associated with strong match-3 games, but expressed through concentric hexagonal geometry and radial collapse rather than vertical gravity.

12. Refill

When no further crunch-generated ring completion exists, remaining empty positions are repopulated with colors from the game's available color inventory. Newly spawned tiles visually scale into place.

In the current prototype, the refill system is mechanically functional but not yet game-directed. It can restore the board, but it does not yet deliberately shape what strategic opportunities, near-completions, difficulty patterns or cascade probabilities should follow.

13. Core Gameplay Loop

7. Read the current distribution of colors across the concentric rings.
8. Navigate the Actor–Follower selection to a useful radial relationship.
9. Swap the two different colors.
10. Observe whether the move improves a ring or completes one.
11. If no ring completes, continue positioning colors.
12. If a ring completes, watch the ring clear and the board crunch inward.
13. Allow any resulting cascade to resolve.
14. Refill remaining empty positions.
15. Read the new board and continue.

The intended puzzle depth comes from choosing swaps that improve future ring formation while understanding how a successful crunch may reorganize the board.

14. Intended Player Experience

14.1 Physical Experience

Playing should feel physically light. The player should be able to hold the phone naturally and play without reaching around the screen. Small thumb gestures should produce clear, substantial visual responses.

14.2 Cognitive Experience

The controls should be learned almost immediately. Difficulty should come from reading the consequences of color redistribution, not from remembering complicated controls.

14.3 Emotional Experience

The main reward moments should be:

- Closing the final gap in a nearly complete ring.
- Seeing an entire monochromatic ring resolve as one coherent structure.
- Watching the board physically crunch inward.
- Receiving an unexpected second or third ring completion from the collapse.

- Feeling responsible for a large chain reaction that originated from one simple swap.

14.4 Mechanical Object Fantasy

Even without a narrative theme, the large rotating hexagon can function as a satisfying object to manipulate. Board rotation, snapping, bouncing, pulling and ring collapse should create a subtle toy-like or machine-like quality.

15. Sources of Puzzle Depth

The game is simple to operate but can become strategically non-trivial because:

- Each ring has a different capacity.
- Every color exists in a finite quantity.
- A color moved inward necessarily leaves its previous position behind.
- A useful swap for one ring may make another ring worse.
- Corner and side positions have different radial connectivity.
- The player can access every valid cross-ring relationship, so difficulty is about decision quality rather than inaccessible moves.
- A completed ring changes more than itself; its removal triggers a structural redistribution of outer colors.
- The crunch can either help or disrupt future ring formation depending on board state and refill behavior.

16. Identity and Distinctiveness

HexCrunch should not be positioned simply as 'a hex match game.' Its identity comes from the combination of several systems:

- A large hexagonal object formed by concentric rings.
- Every neighboring-ring edge is a possible swap relationship.
- A persistent Actor-Follower cursor instead of direct tile tapping.
- One-thumb, low-reach input.
- Board rotation that brings the puzzle to the player.
- Whole-ring monochrome completion rather than local three-piece matching.
- Radial inward collapse instead of vertical gravity.
- Cascade potential generated by geometric crunching.

Many puzzle games contain one or more of these ingredients individually. The design identity lies in how they are composed into a single interaction model and visual experience.

17. Current Design Status

Established / Implemented Core

Open / Not Yet Designed

Concentric-ring hex board geometry	Score system
Actor–Follower selection model	Win condition
One-thumb swipe/tap control	Loss condition
Complete traversal of cross-ring pairs	Level structure
Same-color swap skipping	Difficulty curve
Board rotation/reorientation	Level goals / objectives
Whole-ring monochrome completion	Move limits or time limits, if any
Ring clear	Board-state orchestration / controlled refill
Radial inward crunch	Desired cascade frequency
Color-coherent pull selection	Near-win / near-miss shaping
Cascade-capable repeated resolution	Player progression / meta game
Refill of empty positions	Power-ups / special tiles, if any
Visual states: Idle, Ring Face, Actor, Follower, Blocked, Stray	Monetization model, if any
	Tutorial and onboarding flow
	Long-term retention systems

18. The Next Major System: Board Orchestration

The largest current design limitation is not the lack of a working cascade. The cascade already exists. The missing layer is the logic that intentionally creates good future game states.

A mature HexCrunch board director/refill system may need to reason about:

- How close each ring is to completion.
- Which colors are concentrated or fragmented across rings.
- How many productive moves are available.
- Whether the player is being starved of meaningful choices.
- Whether a refill is setting up a rewarding opportunity, a difficult recovery, or a neutral state.
- How likely a crunch is to generate a secondary or tertiary ring.
- How often spectacular cascades should occur.

- Whether difficulty should come from fragmentation, misleading opportunities, constrained connectivity, or limited resources.
- How to create reward without making outcomes feel scripted or fake.

The guiding principle should be: the player should feel that cascades are earned by intelligent play, even if the game carefully manages the future distribution to preserve fun and pacing.

19. Design Principles to Protect

- Do not turn the game into precision tapping. The 'puzzle comes to the thumb' interaction is a core strength.
- Do not make visual board rotation alter the logical geometry.
- Do not add input steps that communicate no new intention.
- Keep the Actor-Follower relationship legible at all times.
- Let the player understand why a swap is legal or skipped.
- Preserve the sense that the whole hexagon is one physical object.
- Use cascades as a reward amplifier, not as meaningless visual noise.
- Avoid over-scripting refill to the point where player agency feels fake.
- Prefer difficulty from decisions over difficulty from controls.
- Treat sound, haptics, timing and motion as part of gameplay feel, especially during crunches and cascades.

20. Future Design Space (Not Yet Decided)

The following areas are intentionally open. They are design opportunities, not current rules.

- Score based on ring size, cascade depth, efficiency, move quality or speed.
- Level goals such as clearing specified colors, producing specific ring sequences, achieving target cascade depth, or restoring a desired ring arrangement.
- Move-limited puzzle levels versus score-attack or endless modes.
- Difficulty progression through color distribution, refill behavior, altered goals or constrained resources.
- Special events or tiles that modify adjacency, locking, color movement or crunch behavior.
- Combo multipliers for consecutive rings in one cascade.
- Visual/audio escalation as cascade depth increases.
- A solvable-state or target-state mode where the finite 6/12/18/24/30 inventory becomes part of an explicit global sorting objective.
- Different board sizes or ring counts, provided they preserve the tactile clarity of the core system.

21. Glossary

Actor: The selected tile on the inner of the two active neighboring rings.

Follower: The selected adjacent tile on the outer ring; paired with the Actor for a possible swap.

Swap Edge: The radial adjacency relationship represented by the current Actor-Follower pair.

Ring Face: The visual context used to communicate the Actor's ring.

Blocked: A same-color relationship that cannot produce a meaningful swap.

Stray: An empty/off board slot with no color currently assigned.

Crunch: The resolution sequence where a completed ring clears and outer colors are pulled radially inward.

Cascade: One or more additional ring completions generated by the crunch before normal play resumes.

Refill: The process of assigning colors to empty positions after the crunch can no longer generate further ring completion.

Board Orchestration: The future system responsible for intentionally shaping refill and subsequent board states for challenge, fairness, reward and pacing.

22. Design Snapshot

HexCrunch is currently best understood as a one-thumb radial color-sorting puzzle played on a large rotating hexagonal object. The player navigates every valid connection between neighboring concentric rings and swaps colors across those connections. The immediate structural goal is to assemble complete monochromatic rings. A completed ring collapses, causing outer colors to crunch inward through true hexagonal adjacency, with the possibility of chain reactions before refill.

The game's strongest established qualities are its low-friction control scheme, visually distinctive board motion, fixed and learnable geometry, whole-ring objective, and cascade-capable radial resolution. The central unresolved design challenge is how to intelligently direct future board states so that the core system consistently produces meaningful decisions, fair challenge, near-misses and pleasurable cascades over a full game or level progression.

Any future design work should preserve the defining experience: very little physical effort from the player, a large and responsive visual object on screen, transparent and deterministic interaction, and disproportionately satisfying consequences from a small number of simple inputs.